Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

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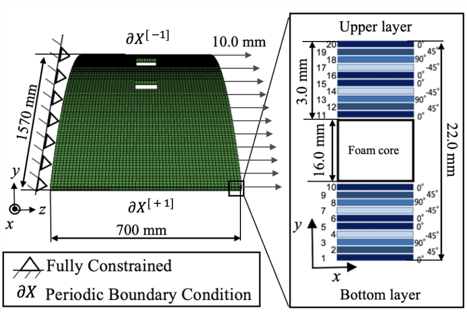
Introduction: CFRP in space transportation

CFRP carries primary load in reusable launch vehicles (high specific strength) [1, 2].

Interlaminar delamination around bolt holes critically degrades integrity, especially under repeated use. Reliable *defect localization* is essential.

Conventional NDT — ultrasonic [3], X-ray CT [4], tap testing [5] — is costly and operator-dependent [6].



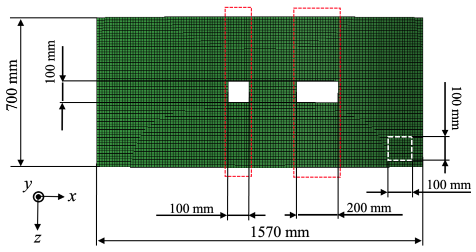
Perforated CFRP interstage panel (FEM).

Infrared stress measurement (DSPSS)

Thermoelastic effect relates a small temperature change to the change in the sum of principal stresses:

$$\Delta T = -k\,T\,\Delta\sigma_{\mathrm{sum}}, \qquad \sigma_{\mathrm{sum}} = \sigma_1 + \sigma_2 + \sigma_3 = \mathrm{tr}(\boldsymbol{\sigma}).$$

Infrared stress measurement [7–9] thus yields a **full-field surface map** of σ_{sum} (DSPSS), from which sub-surface defects are inferred.



DSPSS shows a stress valley over the defect.

Previous study and purpose

- **Previous study** [10, 11]: transfer learning with a CNN estimates defect position/size — but is tied to image grids and struggles on irregular, perforated geometry.
- **This work**: a **Graph Neural Network** (GAT) that consumes FEM-generated DSPSS to predict the *3-D defect location* (in-plane region × insertion layer) in perforated CFRP interstage structures.

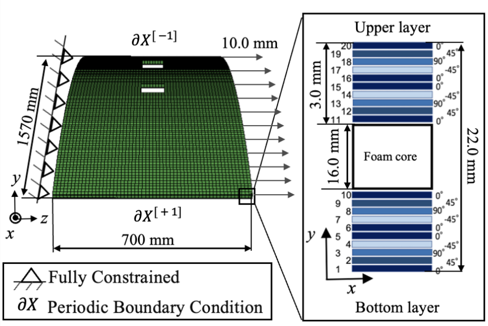
Why a graph?

The FEM surface mesh is naturally a graph; a GNN handles the irregular, hole-perforated topology that a CNN grid cannot.

FEM analysis conditions

- 1/80-scale curved panel ($R \approx 2$ m, $L \approx 1.57$ m, $t = 22$ mm).
- Holes of 100×100 and 200×100 mm.
- Sandwich laminate; tensile loading.
- Interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies.

Synthetic dataset $\Rightarrow$ DSPSS fields with known ground-truth defects.



FEM boundary conditions & hole geometry.

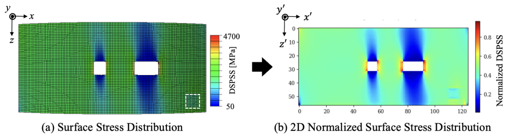
Input data: normalized / difference DSPSS

Raw DSPSS is dominated by **stress concentration around the hole**, masking the defect signal.

Remedy (paper):

- **Difference** field: damaged − defect-free reference;
- **Z-score normalization** (per node), improving depth-direction discrimination.

⇒ robust localization even near the opening.



Normalized DSPSS isolates the defect.

Stress distribution around the defect

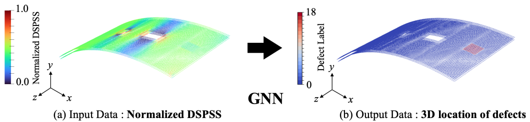
- DSPSS exhibits a clear **stress valley** at the defect centre.
- Depth and width of the valley depend on **ply angle**: 90° plies give sharp narrow valleys; 0°/45° broaden them.
- This ply-angle signature is what the network must learn to separate *which layer* the delamination sits in.

Implication

Layer discrimination is a *fine spatial-frequency* problem — motivating attention over local neighbourhoods.

Graph representation & GAT model

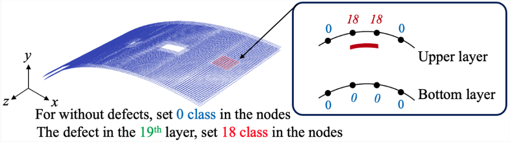
- **Nodes** = FEM surface elements; features $= (x, y, z, \hat{\sigma}_{\text{sum}})$.
- **Edges** = fixed mesh connectivity (reused across graphs).
- **GAT** [12, 13]: attention weights learn neighbour importance; message passing [14] over the mesh.
- Architecture: 3× GATConv (64→256→256), multi-head, batch-norm + dropout + residual connections.



Graph model on the perforated mesh.

Output: 3-D defect location as 19-class node labels

- Node classification into **19 classes**: class 0 = defect-free; remaining classes encode *in-plane region* × *insertion layer*.
- One-side labelling so prediction needs only single-surface DSPSS.



19-class label map.

Severe imbalance

Defect nodes are < 3% of all nodes — handled by the loss (next slide).

Loss function: Focal loss

$$\mathrm{FL}(p_t) = -\,\alpha_t\,(1 - p_t)^{\gamma}\,\log p_t\,.$$

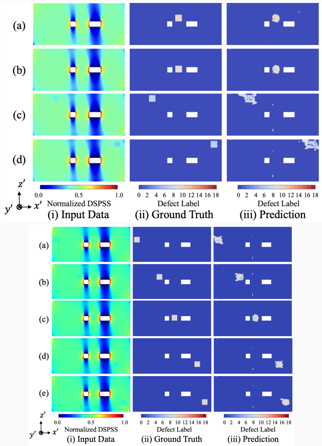
- Down-weights easy, abundant defect-free nodes; emphasises rare defect-class nodes [15].
- Crucial because intact regions dominate ($>97\%$).

Node features normalized; a single fixed edge list reused across all graphs.

Results: prediction accuracy

Test-set metrics:

- planar $R = 0.72$ (mean), depth $P_{\text{gauss}} = 0.69$, combined TDPS $= 0.70$ (best 0.92).
- Best layers: 9–12 (mid-thickness); Layer-10 samples all TDPS ≥ 0.82.
- False-negative rate for defect nodes $< 3\%$.



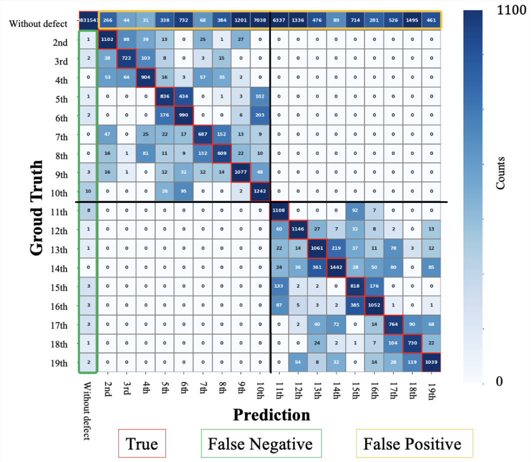
Predicted vs. true defect regions.

Overall performance & confusion

Headline

Macro-F1 = 61% across 19 classes, using *surface DSPSS only*.

- Dominant diagonal; residual *adjacent-layer* confusion.
- False positives cluster near the hole edge.
- Robust across defect insertion regions.



Confusion matrix (macro-F1 61%).

Conclusion

- Introduced an **FEM + GAT** framework for 3-D delamination localization in perforated CFRP interstage panels.
- **Difference-normalized DSPSS** suppresses hole stress concentration and enables full-region estimation.
- Achieved **macro-F1** 61% on 19 classes from surface data only, with a very low defect false-negative rate and no false detections in defect-free regions.

A scalable foundation

toward experimental infrared-stress (TSA) data and practical SHM of aerospace composite structures.

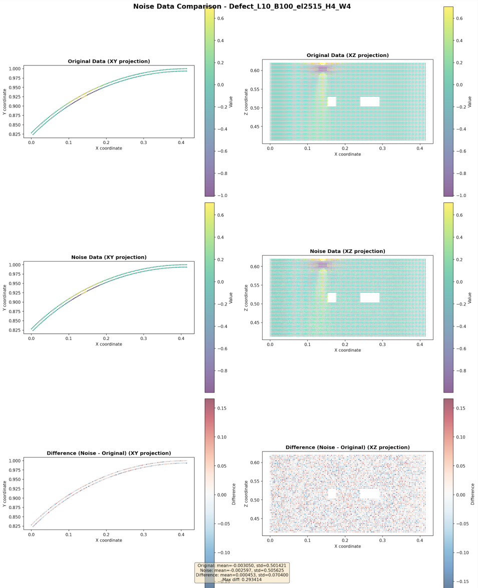
New focus: robustness for real operation

The published model assumes clean FEM fields. In practice:

- infrared stress measurement carries **acquisition noise** (temperature fluctuation, emissivity, sensor);
- the vast majority of a real structure is **defect-free** — false detections there are costly.

WCCM contribution

Train for *robustness under experimentally-motivated noise* and explicitly *suppress false detection* on defect-free regions.



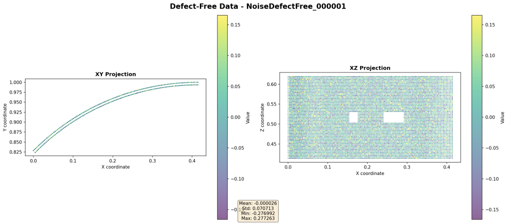
Noise-augmented dataset & defect-free negatives (NDF)

Defect-free negatives (NDF):

- base field set to zero (no defect);
- additive Gaussian noise (10% of data std);
- 5,000 samples added to training.

Noise model on defect data:

- line noise along the fibre (z) direction (random intensity/width);
- attenuated noise in x; global white noise.



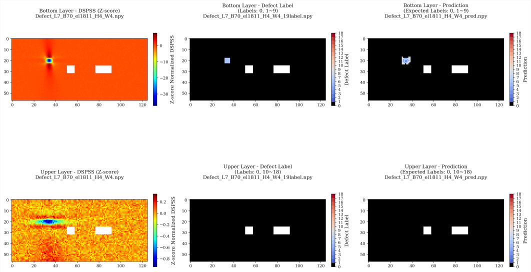
Defect-free noised samples teach the model what "no defect" looks like.

Results under noise

- Difference-DSPSS input enables **full-region** defect estimation.
- Prediction stays accurate on **noise-injected** data.
- **No false detection** on defect-free inputs — the NDF negatives remove spurious activations.

Takeaway

The framework is robust enough to move toward *experimental* infrared stress data.



Prediction on noisy input.

Outlook [ongoing, beyond the published paper]

- **Sim-to-real:** apply the noise-robust model to *experimental* infrared-stress (TSA) measurements.
- **Architecture study (preliminary):** extending from attention GNNs to *mesh-physics* GNNs (MeshGraphNet) and layered variants; early results are encouraging and under evaluation with co-authors.
- **Application:** transfer to aerospace payload-fairing SHM.

Preliminary; not part of the published results.

References

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[10] Y. Kojima et al. **“Inverse estimation method for internal defects based on surface stress of carbon-fiber-reinforced plastics using machine learning”.** In: *Advanced Composite Materials* 31.6 (2022). Published online: 30 Mar 2022, pp. 617–629. DOI: 10.1080/09243046.2022.2052786.

[11] Y. Kojima et al. **“Transfer-learning-aided Defect Prediction in Simply Shaped**

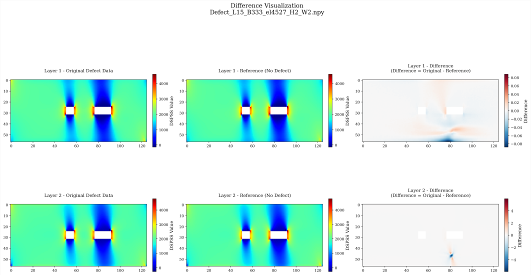
Appendix

Appendix: difference-data construction

Raw DSPSS near a hole is dominated by geometric stress concentration. We remove it by a per-specimen **difference**:

$$\text{Difference} = \underbrace{\text{Original (defect)}}_{\text{damaged}} - \underbrace{\text{Reference (no defect)}}_{\text{pristine}}$$

The residual isolates the defect-induced perturbation, independent of the hole's baseline field.



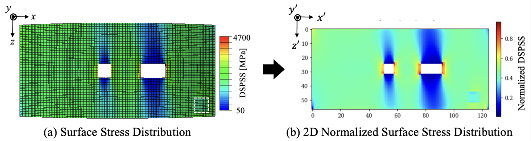
Original − reference = difference field.

Appendix: Z-score normalization

The difference field is standardized per node (Z-score):

$$\hat{\sigma} = \frac{\sigma - \mu}{s}.$$

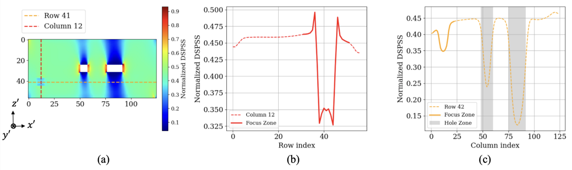
- equalizes magnitude across regions/specimens;
- improves **depth-direction** (layer) discrimination;
- keeps the network NDT-admissible (coordinate/stress only).



Normalized DSPSS input (paper figure).

Appendix: DSPSS profile around the defect

The difference-DSPSS shows a pronounced **stress valley** at the defect centre; its depth and width encode the insertion layer (ply-angle dependence).



Profile of the normalized DSPSS across the defect (paper figure).